Sean Luka Girgis

Senior Python / AWS Data Platform Engineer | PySpark, SQL, Redshift & GenAI

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior Python and AWS Data Platform Engineer with 20+ years of enterprise technology experience across Python, PySpark/Spark, SQL, AWS S3, Glue, Athena, Redshift, EC2/ECS concepts, data modeling, query optimization, Docker, Git, CI/CD concepts, GenAI tools, Bedrock exposure, and financial-services production environments. Strong background building and supporting scalable data pipelines, warehouse-backed reporting structures, forecasting datasets, production validation workflows, and operational dashboards for regulated banking infrastructure. Best aligned to Python/AWS/PySpark data platform roles requiring SDLC discipline, backend/data workflow development, SQL optimization, cloud-native data architecture, GenAI-assisted development, documentation, and cross-functional delivery; NodeJS, DynamoDB, PostgreSQL, Lambda, Step Functions, EventBridge, SNS/SQS, GitLab, Terraform, Postman, and mortgage/bond/trading-domain depth are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Designed and supported Python/Pandas/PySpark and SQL workflows ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Built AWS and warehouse-backed data platform components using S3 landing patterns, AWS Glue, Athena, Amazon Redshift, Oracle, SQL, and PySpark to support scalable ingestion, transformation, querying, and analytics.
- Used complex SQL across Oracle, Amazon Redshift, and reporting structures to validate transformations, troubleshoot discrepancies, tune queries, and improve downstream reporting performance.
- Prepared forecasting-ready datasets for Prophet and scikit-learn workflows, supporting capacity planning, trend analysis, financial-services infrastructure risk review, and bottleneck prediction.
- Implemented data quality and reconciliation checks for row counts, nulls, duplicates, unmatched joins, freshness, abnormal deltas, and output comparisons against trusted prior runs.
- Supported production monitoring and reliability workflows using BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and operational telemetry feeds.
- Created runbooks, source mappings, conceptual data flows, metric definitions, validation checkpoints, support procedures, and technical documentation to improve maintainability and handoff.
- Used AI-assisted development tools including ChatGPT/Copilot-style workflows and built GenAI/Text-to-SQL and RAG-oriented project work to accelerate documentation, code review, test generation, and data workflow development.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Analyzed real-user and synthetic monitoring datasets to identify data quality issues, performance trends, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and performance reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services monitoring and reporting environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing pipelines.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and documentation used by application, middleware, infrastructure, and operations teams.
- Provided technical guidance, onboarding support, training material, and knowledge sharing to improve adoption of monitoring, reporting, and operational standards.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, curated transformations, analytics datasets, query optimization, and validation checkpoints.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

ServiceCall AI RAG Demo
Built staged Python proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, Pydantic schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.
Technologies: Python, FastAPI, Docker, Pydantic, pytest, RAG, TF-IDF

HorizonScale — Capacity Forecasting and Data Platform
Built Python/Pandas/PySpark workflows to process telemetry, validate outputs, prepare forecasting datasets, and produce trusted capacity and risk reporting for financial-services infrastructure.
Technologies: Python, Pandas, PySpark, SQL, Prophet, scikit-learn, Data Validation

AI-Powered Job Search Pipeline
Built an agentic Python pipeline with semantic duplicate detection, LLM job scoring, tailored resume/cover generation, JSON quality gates, and local application tracking.
Technologies: Python, Grok API, FAISS, Sentence Transformers, RAG, JSON, Automation